

Paper A — plan for the October preprint

Written 3 September 2026, after resolving every reference and checking every venue against its official page. Supersedes the positioning in docs/paper-s3-draft.md.

Files: paper/main.tex (skeleton, compiles), paper/references.bib (all resolved), paper/make_figures.py (four figures from archived data), paper/styles/ (official NeurIPS 2026 package, downloaded 3 Sep 2026 from the Call for Papers link).

0. What the reference check changed

Filling the placeholder references forced a read of the full papers. Three claims in the current draft are wrong or overstated, and the paper's positioning changes as a result.

The prior-work map, as verified

paper	date	model(s)	detection measure	content-free vector control	non-introspective prompt	conclusion
Lindsey, <i>Emergent introspective awareness</i>	Oct 2025	Claude Opus 4 / 4.1	generated text, judged	no	no	introspective awareness exists
Godet, <i>Introspection or confusion?</i>	Nov 2025	Mistral-22B, small Qwen/Llama	first-token yes/no logit	informal: "a similar effect is observed with other steering vectors or random ones" (one sentence, no numbers)	no; control questions ("Do you believe 1+1=3?")	steering-induced yes-bias, not introspection
Morris & Plunkett, <i>causal bypassing</i>	Nov 2025	conceptual	—	—	—	detection is the strongest test; identification is confounded
Vogel, <i>Small models can introspect, too</i>	Dec 2025	Qwen2.5-Coder-32B	first-token yes/no logit after "The answer is"	no; control questions	no (all framings introspective; "with info" prompt boosts effect)	detection is real and prompt-dependent
Hahami et al., <i>Detecting the disturbance</i>	Dec 2025	Llama-3.1-8B	binary + localisation + strength discrimination	not stated	no	binary detection confounded by global logit shift; localisation is real
Pearson-Vogel et al., <i>Latent introspection</i>	Feb 2026	Qwen 32B	logit lens across layers	no	no	signal in residual, suppressed at output
Lederman & Mahowald, <i>Content-agnostic</i>	Mar 2026	Qwen3-235B, Llama 405B	generated text judged + logit lens	no (inferred from confabulation patterns: 74.8% "apple")	no; third-person and absurd-question controls	detection is content-agnostic anomaly detection
Macar et al., <i>Mechanisms</i>	Mar 2026	Gemma3-27B , L=37, $\alpha=4$	generated text, GPT-4.1-mini judge	no	partly: "Unprompted" variant, "Notice anything unusual?", no	a post-training circuit implements detection

paper	date	model(s)	detection measure	content-free vector control	non-introspective prompt	conclusion
					preamble; reported as worse (higher FPR, lower TPR)	
Singh, Linzen, Ravfogel, <i>Reality check</i>	May 2026	Llama 70B/8B, Qwen 72B/32B, Gemma-27b-it	first sampled token	no; "gaslight" (input) vs injection	no	sensitivity to generic irregularity
this paper	Sep 2026	Gemma3-27B, L=37	first-token P(YES) and generated text	yes: random + shuffle, norm-matched, pre-registered	yes: preamble-matched neutral prompt	mostly perturbation; small content-specific residual at low α ; not self-specific

Corrections the draft must make

1. **"Critiques attack identification and leave detection standing."** False for Lederman & Mahowald and for Singh et al. Both attack detection's *interpretation* directly. Replace with: two critiques infer generic anomaly sensitivity (from confabulation patterns; from confusion between input and activation manipulation); the defended mechanistic account controls only against no injection.
2. **"The original does not state its normalisation convention."** False. Macar et al. L2-normalise before scaling by α , following Lindsey. Ours are unnormalised (median norm 5,002). Our α is therefore **not** their α , and "10.0% vs 10.8%" is a rate matched by sweeping, not a protocol match. Say so. Run the normalised protocol (§6, run 2).
3. **"The framing control had not been run."** Overstated. Macar et al. have an "Unprompted" variant with no preamble, and found it worse. Ours keeps the preamble and base-rate statement, which is why it has a usable floor. Position as: their variant changed two things at once (framing and preamble); ours changes one.
4. **"The content-free control had not been run."** Overstated by one sentence. Godet (Nov 2025) reports informally that random vectors produce a similar shift on Mistral-22B. Position as: a one-line observation on a different model, never tested systematically, never on the defended model, and not addressed by the two papers that came after it.
5. **"Red apple."** This is Lederman & Mahowald's 74.8% "apple" on a different model. Cite it as a replication of their finding, not a discovery.

Our contribution, restated honestly

- **A direct manipulation where the field had inferences.** Two critiques *infer* that detection is content-agnostic; we *test* it with norm-matched content-free vectors, pre-registered, two independent controls

that agree, on the exact model, layer and prompt of the defended mechanistic account.

- **A qualification neither critique could see.** The shift is *mostly* content-free, not entirely: a concept-specific residual is significant at $\alpha=2$ and 4 ($p \approx 0.001-0.009$ against both controls, under both prompts) and gone by $\alpha=6$. Lederman & Mahowald's method cannot detect a residual; ours puts a number on it.
- **The preamble-matched framing control.** Once anything is injected, the neutral prompt is indistinguishable from the introspective one ($p=0.44-0.75$); the introspective preamble inflates the response to *content-free* vectors eightfold at low strength.
- **The readout swing on one model.** Generated-text detection ranges 7%–50% over strengths where first-token $P(\text{YES})$ is flat. Vogel and Pearson-Vogel show output suppression versus logits; we show the *rate you report* depends on α through the decoding budget.
- **A pre-registered prediction of our own that failed** (the post-hoc-inference account), reported.

The one-line pitch: *the two critiques were right about the bulk of the effect and wrong that it is all of it; the defended paper's own model shows both, once you run the control that separates them.*

1. Venue

Every date below was read from the venue's official page on 3 September 2026.

venue	deadline	status	fit	notes
NeurIPS 2026 · <i>Interpretability as a Science</i>	1 Sep 2026 (AoE)	closed 2 days ago	ideal: "measurement validity, falsifiability, transferable lessons from neuroscience"	Email the organisers today asking whether a late submission is possible. Costs nothing; low odds.
NeurIPS 2026 · Interp4Discovery	3 Sep 11:59 UTC	closes today	weak	—
NeurIPS 2026 · ATTRIB	5 Sep AoE	open	poor: data attribution and provenance	not a fit
NeurIPS 2026 · NeurReps Findings	8 Sep	open	poor: geometry, Nature-calibre experimental collaborations	not a fit
NeurIPS 2026 · UniReps	TBD (tracker says 25 Sep)	uncertain	weak: representational convergence	possible fallback; check unireps.org weekly
arXiv	any	—	—	the date-stamp. Target: first week of October after the runs in §6
ICLR 2027 main	abstract 18 Sep , paper 25 Sep 2026	open	main track; 9 pages	see below
ICLR 2027 workshops	announced ~Dec 2026, deadlines ~Feb 2027	not yet	the natural home	historically several interpretability workshops
ICML 2027 main	abstract 16 Jan, paper 22 Jan 2027	—	Paper B's slot	—
ICML 2027 MechInterp workshop	~May 2027	—	good fit	2026 edition deadline was 8 May
BlackboxNLP 2026	17 Jul 2026	closed	—	BlackboxNLP 2027 ~Jul 2027, backup

Recommendation: arXiv in early October, then an ICLR 2027 workshop in February.

On ICLR 2027 main track. Feasible on paper (3 weeks, 9 pages), but three things argue against it. The work is one model, one layer, 30 concepts — workshop-shaped, and a main-track rejection costs three months of the paper's shelf life. ICLR requires reciprocal reviewing: at least one author must have an accepted paper at a listed top venue, which for a four-student team means the mentor, and that is her decision. And ICLR 2027 makes an **AI-use statement mandatory**, disclosing "significant LLM usage in research ideation and writing." That is a decision for the authors and mentor before any ICLR submission, and it should be made explicitly rather than by default. arXiv has no such requirement; workshops increasingly follow the parent conference's policy, so expect the same question in February.

Workshop format to plan for. NeurIPS/ICLR workshops in this area use the parent style file, 4–6 pages main text excluding references, double-blind, non-archival. The skeleton is built for exactly that: NeurIPS 2026 style, preprint option for arXiv, one line to switch to anonymised.

2. Layout and rules

Style. `neurips_2026.sty`, official package (in `paper/styles/`). For arXiv use `\usepackage[preprint]{neurips_2026}` (authors shown, no line numbers). For a workshop submission use `\usepackage{neurips_2026}` (anonymised, line-numbered) or the workshop's stated option. Never edit the `.sty`.

Length. Target **6 pages** main text for arXiv (workshops cap at 4–6; a 6-page arXiv version cuts to 4 cleanly by moving §3.7 and §3.6 to the appendix). References and appendix unlimited. Currently the skeleton with the four tables and four figures runs about 6.5 pages; trim §1 by a paragraph.

Figures. Full-width figures at 5.5 in (dose-response, paired); half-width at 2.7 in (readout, share). PDF, fonts embedded (`pdf.fonttype 42`). Palette: the three validated categorical slots (blue, orange, aqua) plus distinct markers and dashes so every figure survives greyscale. Check with `pdf fonts main.pdf` that no Type 3 fonts remain.

Tables. `booktabs`, no vertical rules, at most one decimal more than the data supports. Four tables in the main text; the per-concept table goes to the appendix.

Anonymity switch checklist (for the workshop version): remove author block; replace "our earlier work" with third person; strip PDF metadata (`exiftool -all= main.pdf`); no repository URL in the main text, or an anonymised one.

Directory.

<code>paper/</code>	
<code>main.tex</code>	skeleton, compiles now
<code>references.bib</code>	all resolved
<code>make_figures.py</code>	regenerates figures/ from <code>../data/s3/</code>
<code>figures/</code>	<code>fig_dose_response.pdf</code> <code>fig_readout.pdf</code> <code>fig_paired.pdf</code> <code>fig_share.pdf</code>
<code>styles/</code>	<code>neurips_2026.sty</code> <code>neurips_2026.tex</code> <code>checklist.tex</code> (official)
<code>appendix/</code>	(per-concept table, prompt texts, pre-registration excerpts)

Build: `cd paper && pdflatex main && bibtex main && pdflatex main && pdflatex main`.

3. Contents, section by section

Word budgets sum to ~3,200 words of prose, which with four figures and four tables fills 6 pages in NeurIPS format.

§	title	words	what it must do	figures / tables
—	Abstract	220	claim, the two controls, the four numbers, the failed prediction	—
1	Introduction	550	the claim; what a no-injection control cannot separate (three readings); what the two critiques inferred and did not manipulate; what the defended paper controlled for; the two manipulations we add; contributions (5)	—
2	Related work	300	the map in §0 as prose: Lindsey → Godet/Morris/Vogel → Hahami/Pearson-Vogel → L&M / Macar / Singh. One sentence each on what they measured and what they did not. Nisbett & Wilson for confabulation	—
3	Method	550	model, hardware, fp16 overflow; vectors (unnormalised, and why that matters); injection; two prompts verbatim; two readouts; two controls; analysis; pre-registration	Table 1: the two prompts
4.1	Reproduction	120	10.0% [3.5, 25.6] vs 10.8%; 0/30 FPR; now with the caveat that α is matched by rate, not protocol	—
4.2	Readout	180	7%–50% vs flat	Fig. 2 (readout)
4.3	First-token shift; failed prediction	200	0.00003 → 0.417, $p=9.3e-9$; the betrayal quote; outcome A	—
4.4	Content-free controls	350	primary A2 at $\alpha=6$; sweep; 36/61/82%; controls agree; why $\alpha=6$ was the wrong place	Fig. 1 (dose-response), Fig. 3 (paired), Table 2 (sweep)
4.5	Framing	200	indistinguishable once injected; residual survives neutral	Table 3
4.6	Preamble inflation	150	+0.184 vs +0.023	Fig. 4 (share)
4.7	What detections contain	150	0/15 named; 7 echo; red apple → L&M replication; 145×	Table 4
5	Discussion	450	what the no-injection control measures; the residual and the on/off-manifold alternative; "introspective" does no work; readouts; relation to each of the three 2026 papers by name	—
6	Limitations	250	one model/layer/quantisation; trial-number confound and the fix; α scale not theirs ; first-token pooling; one neutral prompt	—
7	Reproducibility	80	free Kaggle, <30 min per condition, version stamp, archived JSONL	—
—	AI-use statement	60	required by ICLR; recommended everywhere. Content is the authors' call	—

§	title	words	what it must do	figures / tables
A	Appendix	—	per-concept P(YES) table ($30 \times 4 \times 3$ vectors $\times 2$ prompts); the four prompt texts; pre-registration excerpts with filing dates; the original scorer vs re-scoring table	Table A1–A3

4. Figures

#	file	shows	status
1	fig_dose_response.pdf	first-token P(YES) vs α , real/random/shuffle, both prompts, 95% bootstrap bands	done
2	fig_readout.pdf	generated-text YES rate vs first-token P(YES) across $\alpha=0\dots 8$	done
3	fig_paired.pdf	per-concept real vs random at $\alpha=2$ and $\alpha=6$, slope chart, 22/30 vs 17/30	done
4	fig_share.pdf	content-free share of the shift, both prompts	done
A1	—	per-concept heatmap of P(YES) (30 concepts $\times 4 \alpha$), real vectors, both prompts	todo, 20 lines
A2	—	the $\alpha=8$ generation collapse: proportion of responses emitting no YES/NO token	todo, 10 lines

All regenerate with `python paper/make_figures.py`; no GPU.

5. Tables

#	content	source
1	the two prompts, verbatim	script constants
2	sweep: real / random / shuffle $\times \alpha$, with both p-values	analyse_s3_full.py
3	introspective vs neutral, real vectors, per α	same
4	what the $\alpha=2$ affirmations contain: named / echo / red apple / other	rescore_s3.py
A1	per-concept first-token P(YES), all cells	data files
A2	run-time scorer vs re-scoring, per α	rescore_s3.py
A3	pre-registration criteria and filing dates	docs

6. Additional runs, ranked

Each is one Kaggle session or less. Total if all run: about 7 GPU-hours, well inside one week's quota. Do them in this order; each one closes a specific objection.

#	run	GPU	closes	changes the paper?
1	L2-normalised vectors, α swept 1–8, introspective + neutral, real + random. Macar et al. normalise; we did not.	~1.5 h	correction 2 above; makes "reproduces their protocol" true or false	yes: §4.1 becomes a protocol match or an honest non-match; may move the concept-specific window
2	Trial-randomised re-run of the four forced conditions (<code>--trial-seed 1</code>)	~2 h	the confound in §6	no numbers change if pairing held; Limitations shortens
3	Second model: Qwen2.5-32B-Instruct, same script, <code>--model qwen</code>	~1.5 h	"one model"	adds a column; turns a finding into a pattern
4	Two more neutral phrasings (e.g. "Is this trial different from the others?", "Has anything been changed?")	~1 h	"one neutral prompt"	strengthens §4.5
5	Second layer (L=25 and L=50, real + random)	~1.5 h	"one layer"	adds an appendix figure
6	Judge check: GPT-4.1-mini on the 210 generated responses vs our rule-based scorer	API cost, no GPU	"rule-based scoring"	replaces or validates Table A2
7	Macar's exact "Unprompted" text as a third framing	~0.5 h	correction 3: lets us compare their variant and ours side by side	strengthens §4.5

Runs 1–3 before arXiv. Runs 4–7 for the workshop version.

Run 1 needs a script change (10 minutes): `--normalise` already exists; add a `--vector-scale` that multiplies unit vectors by a chosen norm, and sweep α on the unit scale. Pre-register the criterion before running: at which α does the normalised protocol reproduce 10.8%, and does real beat random there.

7. Information needed that is not a run

- **Authorship** for a four-person capstone plus mentor. Decide before arXiv; arXiv author lists are changeable but the first version is the one that gets cited.
- **AI-use statement.** Whether and how to disclose the role of LLM tooling in the code, analysis and drafting. Mandatory for ICLR 2027; recommended on arXiv. The authors' call, made explicitly.
- **Lindsey's α definition.** The Transformer Circuits post says "injection strength 2" with no formula; Macar says L2-normalised then scaled. Whether "strength" is relative to the residual-stream norm is still unresolved from the text. Run 1 sidesteps it by sweeping.
- **IFT (Hahami et al. 2026) date.** arXiv ID says July; one index says May. Verify.
- **Gemma 3 report arXiv ID.** Marked "verify" in the .bib.
- **Interpretability-as-a-Science organisers:** one email, today, asking about late submission. Then stop thinking about it.

8. References, resolved

key	status
lindsey2025introspection	resolved; arXiv 2601.01828 (Jan 2026) added
macar2026mechanisms	resolved; full author list; L=37 $\alpha=4$ 10.8%/0% confirmed; L2-normalisation confirmed; "Unprompted" variant found
lederman2026contentagnostic	resolved; changes positioning
singh2026realitycheck	resolved; tests Gemma-27b-it; "generic irregularity"
vogel2025small	resolved; PDF read; control questions not vectors
pearsonvogel2026latent	resolved
godet2025confusion	resolved; contains the one-line random-vector remark
godet2025localization	resolved
morris2025bypassing	resolved
hahami2025disturbance	resolved
hahami2026gift	resolved; date to verify
nisbett1977, wilson1927, wilcoxon1945, dettmers2023qlora	standard
gemma2025gemma3	verify ID

9. Timeline to arXiv

when	what
3–4 Sep	Email Interp-as-Science organisers. Mentor reads <code>PLAN.md</code> §0 and decides on authorship and the AI-use statement.
5–7 Sep	Script change for run 1; pre-register runs 1–3 (one addendum).
8–12 Sep	Runs 1, 2, 3 on Kaggle (~5 GPU-h across the week).
13–17 Sep	Re-run analysis; update tables; rewrite §1, §2, §4.1, §6 for the corrections; port prose into <code>main.tex</code> .
18–22 Sep	academic-paper-reviewer pass (simulated panel); fix what it finds. Mentor read.
23–26 Sep	Final humanizer pass; pdffonts; compile; proofread the PDF, not the source.
~1 Oct	arXiv. cs.LG primary, cs.CL and cs.AI cross-list.
Dec 2026	ICLR 2027 workshops announced; pick one; anonymise; submit Feb 2027.